FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – I

JEE (Main)-2021

TEST DATE: 17-02-2021

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. C

Sol. From $F = \frac{1}{4\pi\varepsilon_0}\frac{e^2}{r^2}$

$\therefore \quad \frac{e^2}{\varepsilon_0} = 4\pi Fr^2$ (dimensionally)

$\frac{e^2}{\varepsilon_0 hc} = \frac{4\pi Fr^2}{hc} = \frac{(MLT^{-2})L^2}{ML^2T^{-1}(LT^{-1})} = [M^0L^0T^0A^0]$,

$\frac{e^2}{\varepsilon_0 hc}$ is called the structure constant and has value $\frac{1}{137}$.

2. C

Sol. Here, $f = f_0\left(1-\frac{t}{T}\right)$ or, $\frac{dv}{dt} = f_0\left(1-\frac{t}{T}\right)$

or, $\quad dv = f_0\left(1-\frac{t}{T}\right)dt$

$\therefore \quad \therefore v = \int dv = \int\left[f_0\left(1-\frac{t}{T}\right)\right]dt$

or, $\quad v = f_0\left(t-\frac{t^2}{2T}\right)+C$

where C is the constant of integration.

At t = 0, v = 0

$\therefore \quad 0 = f_0\left(0 - \frac{0}{2T}\right) + C \Rightarrow C = 0$

$\therefore \quad v = f_0\left(t - \frac{t^2}{2T}\right)$

If f = 0, then

$0 = f_0\left(1 - \frac{t}{T}\right) \Rightarrow t = T$

Hence, particle's velocity in the time interval t = 0 and t = T is given by

$$v_x = \int_{t=0}^{t=T} dv = \int_{t=0}^{T}\left[f_0\left(1-\frac{t}{T}\right)\right]dt$$

$$= f_0\left[\left(t - \frac{t^2}{2T}\right)\right]_0^T$$

$$= f_0\left(T - \frac{T^2}{2T}\right) = f_0\left(T - \frac{T}{2}\right) = \frac{1}{2}f_0T$$

3. C

Sol. Maximum force by surface when friction works

$F = \sqrt{f^2 + R^2} = \sqrt{(\mu R)^2 + R^2} = R\sqrt{\mu^2 + 1}$

Minimum force = R when there is no friction

Hence ranging from R to $R\sqrt{\mu^2 + 1}$ [where, R = mg]

4. D

Sol. The velocity of particle after falling through height h

$u = \sqrt{2gh}$ …(i)

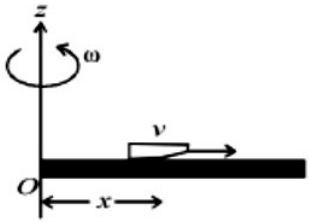
After first rebounding, the velocity of ball is eu and after attaining maximum height it will come to the ground with same velocity eu. So, after second rebounding its velocity will be e^2u.
Similarly, after third fourth …etc. rebounding it velocities will be e^2u, e^4u, … etc.

5. B

Sol. We know that $|\vec{\tau}| = \left|\frac{d\vec{L}}{dt}\right|$, where $L = I\omega$

$\therefore \quad \tau = \frac{d}{dt}(I\omega) = \omega\frac{dI}{dt}$ …(i)

From the situation it is clear that the moment of inertia for (rod + insect) system is increasing.



Let at any instant of time 't', the insect is at a distance x from O. At this instant, the moment of inertia of the system is

$I = \frac{1}{3}ML^2 + mx^2$ …(ii)

From (i) & (ii)

$$\tau = \omega\frac{d}{dt}\left[\frac{1}{3}ML^2 + mx^2\right] = \omega m\frac{d}{dt}(x^2)$$

$$= 2\omega mx\frac{dx}{dt} = 2\omega mxv = 2\omega mv^2t$$

$\therefore$ $\tau \propto t$ (till t = T)

When the insect stops moving, $\vec{L}$ does not change and therefore τ becomes constant.

6. B

Sol. If T be the tension in the rope, then 2T = 10 kN

2T = 10kN

T = 5kN

$\therefore$ Longitudinal stress in the rope

$$\sigma = \frac{T}{A} = \frac{5kN}{10^3 mm^2} = 5\,Nmm^{-2}$$

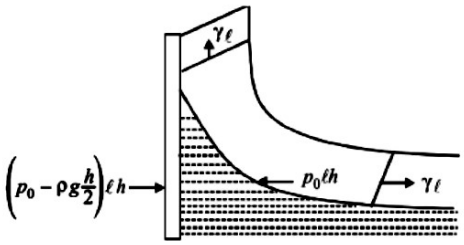
$\therefore$ Extension in the rope

$$= \frac{Stress}{Y}\times L = \frac{5\,Nmm^{-2}}{10^3 Nmm^{-2}}\times 1500mm = 7.5mm$$

$\therefore$ Deflection of the load $\delta = \frac{7.5}{2} = 3.75mm$

7. A

Sol.



Balancing forces in horizontal direction

$$\left(p_0 - \rho g\frac{h}{2}\right)\ell h + \gamma\ell = p_0\ell h \Rightarrow h = \sqrt{\frac{2\gamma}{\rho g}}$$

8. D

Sol. Since tension in the two rods will be same, hence

$$A_1Y_1\alpha_1\Delta\theta = A_2Y_2\alpha_2\Delta\theta$$

$\Rightarrow$ $A_1Y_1\alpha_1 = A_2Y_2\alpha_2$

9. D

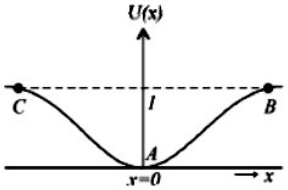
Sol. The heat capacity of the solid is

$$\frac{dQ}{d\theta} = 2a\theta + 4b\theta^3$$

10. D

Sol. Let us plot the graph of the mathematical equation

$$U(x) = K\left[1 - e^{-x^2}\right], F = -\frac{dU}{dx} = 2kxe^{-x^2}$$



From the graph it is clear that the potential energy is minimum at x = 0. Therefore, x = 0 is the state of stable equilibrium. Now if we displace the particle from x = 0 then for displacement the particle tends to regain the position x = 0 with a force $F = \frac{2kx}{e^{x^2}}$. Therefore for small values of x we have $F \propto x$.

11. D

Sol. $A_A = 2, f_A = f$

and $A_B = 1, f_B = 2f$

$\therefore \quad \frac{I_A}{I_B} = \frac{f_A^2 A_A^2}{f_B^2 A_B^2} = 1$

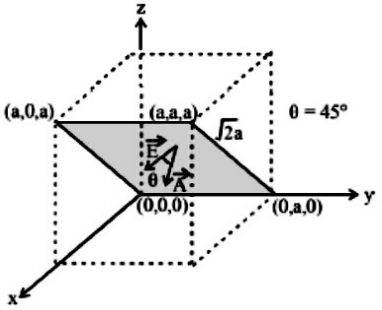
12. C

Sol. Given $\vec{E} = E_0 \hat{x}$

This shows that the electric field acts along +x direction and is a constant. The area vector makes an angle of 45° with the electric field. Therefore the electric flux through the shaded portion whose area is $a \times \sqrt{2}a = \sqrt{2}a^2$ is

$\phi = \vec{E}.\vec{A} = EA\cos\theta = E_0(\sqrt{2}a^2)\cos 45° = E_0(\sqrt{2}a^2)$

$\times \frac{1}{\sqrt{2}} = E_0 a^2.$



13. D

Sol. Fundamental frequency, $f = \frac{1}{2L}\sqrt{\frac{T}{\mu}}$

T = force on dielectric slab.

To calculate force on slab, consider capacitor as combination of two capacitors one with slab and one with air.

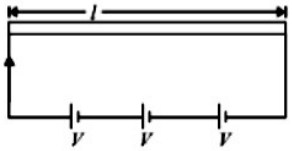
$F = -\frac{dU}{dx}; U = \frac{1}{2}\left[\frac{k \in_0 bx}{d} + \frac{\in_0 b(b-x)}{d}\right]V^2$

$\frac{dU}{dx} = \frac{1}{2}\left[\frac{k \in_0 b}{d} - \frac{\in_0 b}{d}\right]V^2 = \frac{1}{2}\frac{\in_0 bV^2}{d}(k-1)$

Thus, $f = \frac{1}{2L}\sqrt{\frac{\in_0 bV^2(k-1)}{2d\mu}}$

14. B

Sol. Let R be the resistance of wire.



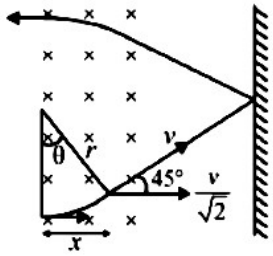
Energy released in t second $= \frac{(3V)^2}{R} \times t$

$\therefore \quad Q = \frac{PV^2}{R} \times t$ But Q = mcΔT (where m = mass of wire)

$\therefore \quad mc\Delta T = \frac{9V^2}{R} \times t$

15. A

Sol. $r = \frac{mv}{qB}$



$\sin\theta = \frac{x}{r}$

$= \frac{\frac{mv}{\sqrt{2}qB}}{\frac{mv}{qB}} = \frac{1}{\sqrt{2}}$

or $\theta = \frac{\pi}{4}$

Time to complete the circle (2π), $T = \frac{2\pi m}{qB}$

$\therefore$ Time taken to traverses $\frac{\pi}{4}, t = \frac{\pi m}{4qB}$

$t_1 = \frac{\frac{mv}{\sqrt{2}qB}}{\frac{v}{\sqrt{2}}} = \frac{m}{qB}$

Total time taken = $2t + 2t_1$

$= \frac{m}{2qB}(\pi + 4)$

16. D

Sol. $e = -\frac{d\phi}{dt} - A\frac{dB}{dt} = -\pi r^2 \frac{d}{dt}(B_0 e^{-t})$

$= \pi r^2 B_e e^{-t}$

At $t = 0, e = \pi r^2 B_0$

Power developed, $P = \frac{e^2}{R} = \frac{(\pi r^2 B_0)^2}{R} = \frac{\pi^2 r^4 B_0^2}{R}$

17. D

Sol. Impedance (Z) of the series LCR circuit is

$Z = \sqrt{R^2 + (X_L - X_C)^2}$

At resonance, $X_L = X_C$

Therefore, $Z_{minimum} = R$

18. C

Sol. **Given:** Amplitude of electric field,
$E_0 = 4$ v/m
Absolute permittivity,
$\varepsilon_0 = 8.8 \times 10^{-12} c^2 N - m^2$
Average energy density $u_E = ?$
Applying formula,
Average energy density $u_E = \frac{1}{4}\varepsilon_0 E^2$

$\Rightarrow \quad u_E = \frac{1}{4} \times 8.8 \times 10^{-12} \times (4)^2$

$= 35.2 \times 10^{-12} J/m^3$

19. A

Sol. Let refractive index of glass be μ.
Let after first refraction, image distance b v then

$$\frac{\mu}{v} - \frac{1}{\infty} = \frac{\mu - 1}{R} \Rightarrow v = \frac{\mu R}{\mu - 1}$$

Now second refraction will take place.
So, distance of first image from O is

$$u_1 = \frac{\mu R}{\mu - 1} - R = \frac{R}{\mu - 1}$$

and image is formed at R

$$\therefore \frac{1}{R} - \frac{\mu(\mu - 1)}{R} = \frac{2(1 - \mu)}{R}$$

$\Rightarrow \quad \mu^2 - 3\mu + 1 = 0$ So, $\mu = \frac{3 + \sqrt{5}}{2}$

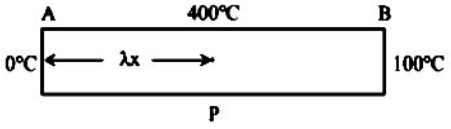
20. C

Sol. Facts

SECTION – B

21. 00009.00

Sol.



For heat flow from P to 0

$$L_f \frac{dm_1}{dt} = \frac{KA400}{\lambda x} \qquad \ldots(i)$$

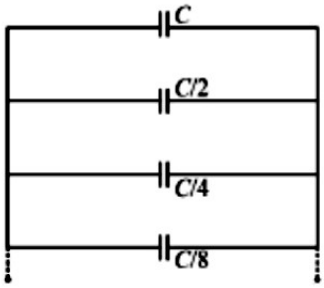
For heat flow from P to B

$$L_{vap} \frac{dm_2}{dt} = \frac{KA300}{10x - \lambda x} \qquad \ldots(ii) \left[\text{Given} \frac{dm_1}{dt} = \frac{dm_2}{dt} \right]$$

On solving (i) and (ii), we get $\lambda = 9$

22. 00002.00

Sol. The equivalent system of capacitors is shown in figure.



If C is the equivalent capacitance, then

$$C' = C + \frac{C}{2} + \frac{C}{4} + \frac{C}{8} + ...$$

$$= C\left[\frac{1}{1-\frac{1}{2}}\right] = 2C = 2\mu F$$

23. 00025.00

Sol. We have $K_\alpha = \frac{m_y}{m_y + m_\alpha}.Q$

$$\Rightarrow \qquad K_\alpha = \frac{A-4}{A}.Q \Rightarrow 48 = \frac{A-4}{A}.50 \Rightarrow A = 100$$

24. 00002.00

Sol. For hydrogen like atom energy of the nth orbit is

$$E_n = -\frac{13.6}{n^2}Z^2 \text{ eV/atom}$$

For transition from n = 5 to n = 4,

$$hv = 13.6 \times 9\left[\frac{1}{16} - \frac{1}{25}\right] = \frac{13.6 \times 9 \times 9}{16 \times 25} = 2.754 \text{ eV}$$

For transition from n = 4 to n = 3,

$$hv' = 13.6 \times 9\left[\frac{1}{9} - \frac{1}{16}\right] = \frac{13.6 \times 9 \times 7}{9 \times 16} = 5.95 \text{ eV}$$

For transition n = 4 to n = 3, the frequency is high and hence wavelength is short.

For photoelectric effect, $hv' - W = eV_0$, where W = work function

$5.95 \times 1.6 \times 10^{-19} - W = 1.6 \times 10^{-19} \times 3.95$

$\Rightarrow \qquad W = 2 \times 1.6 \times 10^{-19} = 2 \text{ eV}$

25. 00015.00

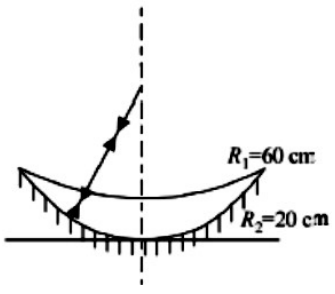
Sol. This silvered concavo-convex lens behaves like a mirror whose focal length can be calculated by the formula

$$\frac{1}{f} = \frac{2}{f_1} + \frac{1}{f_2}$$

f_1 = focal length of concave surface.

f_2 = focal length of concave mirror.

$$\therefore \qquad \frac{1}{f} = \frac{2}{-30} + \frac{1}{-10} = \frac{4}{30}$$



$\therefore$ f = –7.5 cm

Using mirror formula

$$\frac{1}{f}=\frac{1}{v}+\frac{1}{u}\Rightarrow\frac{1}{-7.5}=\frac{1}{-x}+\frac{1}{-x}\Rightarrow x=15\text{ cm}$$

26. 00012.50

Sol. Circuit can be redrawn as figure (A)

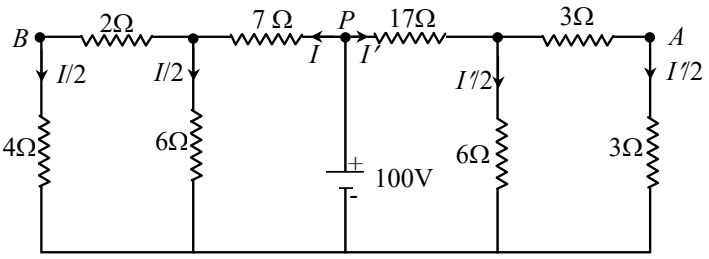


Figure (A)

It is further reduced using series and parallel combination of resistances and is shown in figure (B) and consequently in figure (C)

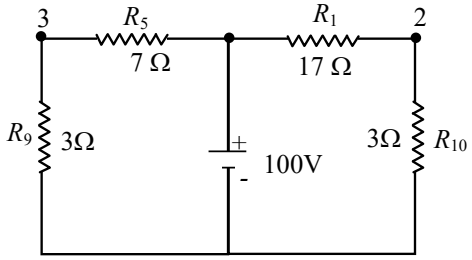


Figure (B)

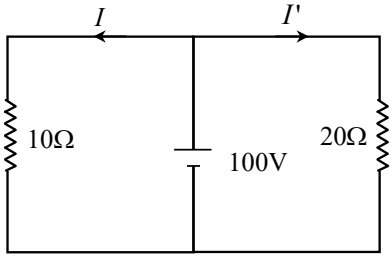


Figure (C)

Current in the 10 Ω resistor is $I=\frac{100}{10}=10\,\text{A}$ and in the 20 Ω resistor is $I'=\frac{100}{20}=5\,\text{A}$

As shown in figure (A) potential difference between points *P* and *B* is

$V_P-V_B=10\times7+5\times2$ =80V … (i)

Also potential difference between points *P* and *A* is

$V_P-V_A=5\times17+2.5\times3$ = 92.5 V … (ii)

Subtracting (i) and (ii) we get, $V_B-V_A=12.5$ V

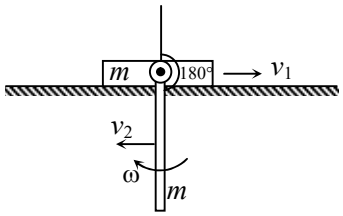
27. 00010.00

Sol. There is no horizontal force, momentum is conserved

$mv_1-mv_2=0$

For hinged point

$v_1=\frac{L}{2}\omega-v_2$



Energy conservation, $mgL=\frac{1}{2}mv_1^{\,2}+\frac{1}{2}mv_2^{\,2}+\frac{1}{2}\frac{mL^2\omega^2}{12}$

Solving, $v_1=\sqrt{\frac{3}{5}gL}$, v_1 = 10 m/s.

28. 00001.33

Sol. Use refraction formula at both surfaces

29. 00020.00

Sol. $\vec{V}_{\text{obj, mirror}} = 4\hat{i} + 9\hat{j}$

$$\frac{dx}{dt} = 4, \frac{dy}{dt} = 9$$

$$u = -x$$

$$-\frac{1}{10} = \frac{1}{V} + \frac{1}{-x}$$

$$V = \frac{-10x}{x-10}$$

$$V_{1x} = \frac{dV}{dt} = \left(\frac{10}{x-10}\right)^2 \left(\frac{dx}{dt}\right)$$

$$V_{1x} = -16$$

$$m = -\frac{V}{-x} = \frac{-10}{x-10} = \frac{y_1^0}{y}$$

$$V_{1y} = -\left(\frac{10}{x-10}\right)\frac{dy}{dt}$$

$$V_{1y} = -18$$

$$\vec{V}_{\text{image, mirror}} = -16\hat{i} - 18\hat{j}$$

$$\vec{V}_{\text{image}} = -12\hat{i} - 16\hat{j}$$

$$|V| = 20 \text{ cm/s}$$

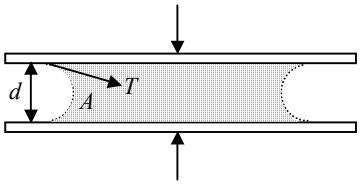
30. 00006.40

Sol. Let R be the radius of the circular layer of water.

Then, $\pi R^2 d \times \rho = m$ …(1)

Pressure at $A = p_0 - \frac{2T}{d}$ …(2)

Thus pressure between the plates is less than the atmospheric pressure and so the plates are pressed together as though attracted towards each other.



F, force of attraction = $\Delta p \times$ area $\Rightarrow F = \frac{2T}{d} \times \pi R^2$

$$\Rightarrow \quad F = \frac{2T}{d} \times \frac{m}{d \cdot \rho} = \frac{2Tm}{d^2 \cdot \rho} = \frac{2 \times 0.4 \times 10^{-3} \times 0.08}{0.01^2 \times 10^{-4} \times 1000} = 6.4 \text{ N}$$

Chemistry

PART – II

SECTION – A

31. B

Sol. $2NH_3(g) + 2O_2(g) \rightleftharpoons N_2O(g) + 3H_2O(g)$

$$K_c = \frac{(N_2O) \times (H_2O)^3}{(NH_3)^2 \times (O_2)^2}$$

32. B

Sol. $$\Delta S = nR\ln\left(\frac{V_f}{V_i}\right) = R\ln\left(\frac{P_i}{P_f}\right)$$

$$R\ln\left(\frac{300R}{1L \times 1atm}\right) = R\ln(24.6)$$

33. A

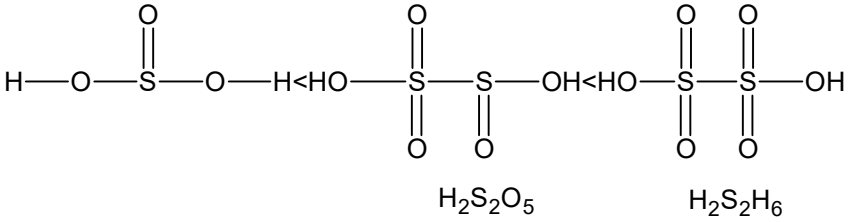
Sol. Bond angle of group 15 hydrides is in order of

$SbH_3 < AsH_2 < PH_3 < NH_3$

91.3^o 91.8^o 93.8^o 104.2^o

34. B

Sol. H_2SO_3



$H_2S_2O_5$ $H_2S_2H_6$

35. C

Sol. Factual

36. C

Sol. Fe^{2+} $n = 0$ $MM = 0$

Fe^{2+} $n = 4$ $MM = \sqrt{24}$

37. A

Sol. $X^{2+} + 2e^- \rightarrow X; Y^+ + e^- \rightarrow Y$

2 mol e^- produce X = 1 mol = $1 \times$ M g

2 mol e^- produce Y = 2 mol = $2 \times$ M/2=M g

38. D

Sol. (A) $Fe^{3+} + 3SCN \rightarrow Fe(SCN)_3$ (red solution)

(B) Red solution of iron (II) dimethylglyoxime.

(C) $Hg^{2+} + 2I^+ \rightarrow HgI_2 \downarrow$ (red).

39. B

Sol. For Lyman series $n_1 = 1$

For shortest 'λ' of Lyman series the energy difference in two levels showing transition should be maximum (i.e. $n_2 = \infty$).

$$\frac{1}{\lambda} = R_H\left[\frac{1}{1^2} - \frac{1}{\infty^2}\right] = 109678$$

$$\therefore \quad \lambda = 911.7 \times 10^{-8} = 911.7\,\text{Å}$$

40. C

Sol. $$n_{H_2} = \frac{2g}{2g/mol} = 1\,mol. n_{SO_2} = \frac{32g}{64g/mol} = 0.5mol$$

$$\therefore P_{H_2} = \frac{n_{H_2}}{\left(n_{H_2} + n_{SO_2}\right)} \times P_T = \frac{1}{(1+0.5)} \times P_T = \frac{2}{3}P_T$$

41. D

Sol. $$\text{Neucleophilicity} \propto \text{Tendency to Donate } \bar{e} \propto \frac{1}{\text{stability of anion}}$$

42. D

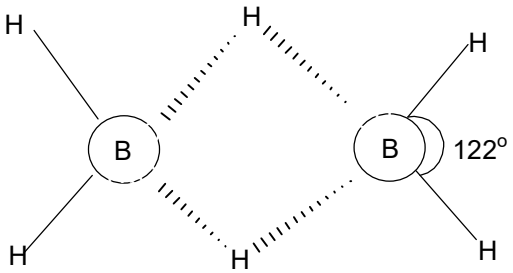
Sol. $HCl \rightarrow H$ has + 1 oxidation state so it resemble 1A group property

$NaH \rightarrow H$ has -1 oxidation state so it resemble VII A group property

$H(1s^1)$ due to half filled shell of electron it resemble IV A group property.

43. C

Sol.



Three centered bond. Called banana bond.

44. D

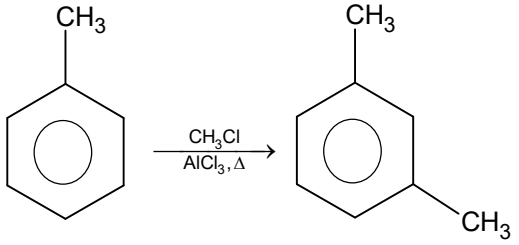
Sol. $$C_6H_5 - \underset{\underset{O}{||}}{C} - H + CH_2 - \underset{\underset{O}{||}}{\underset{\underset{H}{|}}{C}} - H \rightarrow C_6H_5 - \underset{\underset{OH}{|}}{CH} - CH_2 - \underset{\underset{O}{||}}{C} - H.$$

L.H. absent Aldol

In this reaction only one kind of cross is formed, because Benzaldehyde can't be have αH. Therefore option (D) is correct.

45. D

Sol.



m-xylene is thermodynamically more stable than others.

46. C

Sol. The letter D in D-glucose signifies its relative configuration, therefore option (C) is correct.

47. B

Sol. anode: $Ag(s) \rightarrow Ag^{+}(aq) + 1e^{-}$

cathode: $Ag^{+}(aq) + 1e \rightarrow Ag$

net: $Ag^{+}_{(AgBr)} \xrightarrow{1e^{-}} Ag^{+}_{(Ag_2CO_2)}$

$$0 = 0 + \frac{0.059}{1}\log\frac{\left(\frac{K_{SP}AgBr}{[Br^{-}]}\right)}{\sqrt{\frac{K_{SP}Ag_2CO_3}{\left[CO_3^{2-}\right]}}}$$

$$\Rightarrow \frac{K_{SP}AgBr}{\left[Br^{-}\right]} = \sqrt{\frac{K_{SP}Ag_2CO_3}{[CO_3^{2-}}}$$

48. B

Sol. For Lyman series $n_1 = 1$

For shortest 'λ' of Lyman series the energy difference.

49. A

Sol. $$K_b = \frac{x^2}{1-x} = 2.4\times10^{-2} = K_b$$

$$x^2 + (2.4\times10^{-2})x - 2.4\times10^{-2} = 0$$

$$x = 0.143 = [OH^{-}]$$

$$[H^{+}] = \frac{K_w}{[OH^{-}]} = 7\times10^{-14}M$$

$$\therefore pH = 13.15$$

50. A

Sol. Hints: $\frac{\text{Volume of } H_2}{\text{Volume of } O_2} = \frac{\text{Equivalent volume of } H_2}{\text{Equivalent volume of } O_2}$.

SECTION – B

51. 00003.00

Sol. $E = \frac{m}{Q} \times 96500 = \frac{22.2}{2 \times 5 \times 60 \times 60} \times 96500 = 59.5$

$\text{Oxi. state} = \frac{\text{At.wt.}}{\text{Eq.wt.}} = \frac{177}{59.5} = 3$

52. 00002.00

Sol. E_1 for H = -13.6V

$\therefore E_2$ for H = $(-13.6/2^2) = -13.6/4 = -3.4\text{eV}$

$\therefore E_2 - E_1 = -3.4 - (-13.6) = +10.2\text{eV}$

Also for transition of H like atom; $\lambda = 3.0 \times 10^{-3}\text{m}$

$\frac{1}{\lambda} = R_{H^+} Z^2 \left[\frac{1}{1^2} - \frac{1}{2^2}\right]$

$= \frac{1}{3 \times 10^{-8}} = 1.09 \times 10^7 \times Z^2 \times \frac{3}{4}$

$\therefore Z^2 = 4$ and Z = 2

53. 00015.54

Sol. Let there is xFe atom in +3 state

$3x + 2(93 - x) = 200$

$x = 14$

$\%Fe^{3+} = \frac{14}{93} \times 100\% = \frac{1400}{93}\%$

≈ 15.54

54. 00014.30

Sol. Mole of KOH = $\frac{5.6}{56} = 0.1$

$[OH] = 2M$

$pOH = -0.3 \Rightarrow pH = 14.3$

55. 00014.00

Sol. $[OH] = 1m$

$pOH = 0 \quad \Rightarrow pH = 14$

56. 00001.20

Sol. $d = M\left(\frac{1}{m} + \frac{\text{mol.wt.}}{1000}\right) = 11.12\left(\frac{1}{94.13} + \frac{98}{1000}\right) = 1.20\text{g/ml}$

57. 00000.09

Sol. $\frac{VO_2}{V_{total}} = \frac{\frac{4}{3}\pi(2 \times 10^{-10})^3 \times N_A \times 100}{22.4 \times 10^{-3}}$

$= 0.09\%$

58. 00025.00

Sol. $\therefore$ moles dissociates = 100-75 = 25

$$\% \text{ Dissociation } = \frac{25}{100} \times 100 = 25\%$$

59. 00038.70

Sol. $9.3 = k_r \times m$

$$m = \frac{9.3}{1.86} = 5(\text{M.wt} = 62)$$

$$5 = \frac{(50/62)}{wt_{solvent}} \times 1000$$

$$Wt_{solvent} = \frac{50}{62} \times 1000$$

= 161.3 g

$$1Cl = 200 - 161.3$$

$$= 38.7g$$

60. 00000.75

Sol. $$= \frac{1}{75} \times \frac{1000}{18} = 0.75$$

Mathematics

PART – III

SECTION – A

61. B

Sol. Let $S = \sum_{n=2}^{\infty} \log_2\left(1-\frac{1}{n^2}\right) = \sum_{n=2}^{\infty}\left(\log_2\left(1-\frac{1}{n}\right)+\log_2\left(1+\frac{1}{n}\right)\right) = \sum_{n=2}^{\infty}\left(\log_2\left(\frac{n-1}{n}\right)-\log_2\left(\frac{n}{n+1}\right)\right)$

Let $V_r = \log_2\left(\frac{r-1}{r}\right) \Rightarrow V_{r+1} = \log_2\left(\frac{r}{r+1}\right)$

$\Rightarrow S = \sum_{r=2}^{\infty}(V_r - V_{r+1}) = V_2 = \log_2\left(\frac{1}{2}\right) = -1$

62. B

Sol. We have $A^2 = (AB)A = A(BA) = AB = A$

Similarly $B^2 = B$: $(A + B)^2 = A^2 + B^2 + AB + BA = A + B + A + B = 2(A + B)$

$\Rightarrow |A + B|^2 = 8|A + B| \Rightarrow |A + B| = 8$ and $(A – B)^2 = A^2 + B^2 – AB – BA = A + B – A – B = O$

$\Rightarrow |A – B| = 0$

63. A

Sol. For $n > 1$, 2^n is a multiple of 4

Let $2^n = 4k$;

Digit at units place in 2^4 is 6 and the digit at units place in positive integral power of 6 is 6.

The digit at units place in 5!, 6! ….. 20! is zero and also 0! + 1! + 2! + 3! + 4! = 34

$\therefore$ Digit at units place = 6 + 4 = 10

$\Rightarrow$ '0'

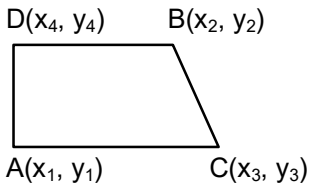
64. A

Sol. On solving we have

$(x_1 - x_3)^2 + (x_2 - x_4)^2 + (y_2 - y_3)^2 + (y_1 - y_4)^2 \le 0$

$\Rightarrow x_1 = x_3,\ x_2 = x_4,\ y_2 = y_3$ and $y_1 = y_4$

$\Rightarrow \frac{x_1 + x_2}{2} = \frac{x_3 + x_4}{2}$ and $\frac{y_1 + y_2}{2} = \frac{y_4 + y_3}{2}$



Hence AB and CD bisect each other

Also we have $AB^2 = CD^2$

$\Rightarrow$ AB = CD

65. B

Sol. Since origin is centre of given circle and in an equilateral triangle centroid and circumcentre are same point

$\Rightarrow \frac{a}{3}\left(\sum \cos\theta_i\right) = \frac{a}{3}\left(\sum \sin\theta_i\right) = 0$

66. A

Sol. On solving we have $\frac{y}{x} = \log x - \log(a + bx)$

$\Rightarrow \frac{x\frac{dy}{dx} - y}{x^2} = \frac{1}{x} - \frac{b}{a+bx}$

$\Rightarrow x\frac{dy}{dx} - y = \frac{ax}{a+bx} \Rightarrow x\frac{d^2y}{dx^2} + \frac{dy}{dx} - \frac{dy}{dx} = \frac{a(a+bx) - abx}{(a+bx)^2}$

$\Rightarrow x^3\dfrac{d^2y}{dx^2}=\dfrac{a^2x^2}{(a+bx)^2}=\left(x\dfrac{dy}{dx}-y\right)^2$

67. B

Sol. Slope of the line = $\dfrac{b}{a}\left[\dfrac{\sin 60^\circ-\sin 30^\circ}{\cos 60^\circ-\cos 30^\circ}\right]=-\dfrac{b}{a}=-\dfrac{1}{2}\Rightarrow a=2b$

Homogenizing the ellipse with $x+2y+4=0$, we get

$\dfrac{x^2}{4b^2}+\dfrac{y^2}{b^2}=\left(\dfrac{x+2y}{-4}\right)^2$

Again coefficient of x^2 + coefficient of $y^2 = 0$

$\Rightarrow \dfrac{1}{4b^2}-\dfrac{1}{16}+\dfrac{1}{b^2}-\dfrac{4}{16}=0\Rightarrow b^2=4\Rightarrow b=2$ and $a=4$

68. B

Sol. Given $\dfrac{c+1}{c-1}=\dfrac{(x^2+x+1)^2}{x^4+x^2+1}=\dfrac{(x^2+x+1)^2}{(x^2+x+1)(x^2-x+1)}=\dfrac{x^2+x+1}{x^2-x+1}$

$\Rightarrow c=\dfrac{x^2+1}{x}=x+\dfrac{1}{x}=f(f(x))+f\left(f\left(\dfrac{1}{x}\right)\right)$

$\because \left[f(f(x))=\dfrac{1-f(x)}{1+f(x)}=\dfrac{1-\dfrac{1-x}{1+x}}{1+\dfrac{1-x}{1+x}}=x\right]$

69. A

Sol. Let W = {CAT, TOY, YOU, ...}

Clearly, R is reflexive and symmetric but not transitive

Since CAT R TOY, TOY R YOU $\not\Rightarrow$ CAT R YOU

70. A

Sol. As $\tan\alpha = 2\alpha$ and $\tan\beta = 2\beta$

$\Rightarrow 2(\alpha+\beta)=\dfrac{\sin(\alpha+\beta)}{\cos\alpha\cos\beta}$ and $2(\alpha-\beta)=\dfrac{\sin(\alpha-\beta)}{\cos\alpha\cos\beta}$

Now $\displaystyle\int_{-1}^{1}\left[\cos(\alpha-\beta)x-\cos(\alpha+\beta)x\right]dx=\dfrac{\sin(\alpha-\beta)}{\alpha-\beta}-\dfrac{\sin(\alpha+\beta)}{\alpha+\beta}=0$

71. B

Sol. Number of one–one onto mapping having exactly 3 elements having self image = ${}^5c_3\cdot 1=10$

Number of one–one onto mapping having exactly 5 elements having self image = 1

$\therefore$ Required number = 10 + 1 = 11

72. D

Sol. As $f(x)g(x)=2$ ..... (a) $\Rightarrow f'(x)g(x)+f(x)g'(x)=0$ ..... (1)

$\Rightarrow f''(x)g(x)+2f'(x)g'(x)+f(x)g''(x)=0$

$\Rightarrow \dfrac{f''(x)g(x)}{f(x)g'(x)}+\dfrac{2f'(x)}{f(x)}+\dfrac{g''(x)}{g'(x)}=0$ (Dividing throughout by $f(x)\,g'(x)$)

$\Rightarrow \dfrac{f''(x)}{f'(x)}-\dfrac{g''(x)}{g'(x)}=\dfrac{2f'(x)}{f(x)}$ (Replacing $\dfrac{g(x)}{f(x)g'(x)}=-\dfrac{1}{f'(x)}$ from (1))

$\frac{2f'(x)}{f(x)} = 0$ has no real roots such $f(x) \neq 0$ for any x (from (a))

73. C

Sol. We know that if $d_i = \frac{x_i - A}{h}$ then $\sigma_x = |h|\sigma_d$

In this case $-2x_i - 3 = \frac{x_i + \frac{3}{2}}{-\frac{1}{2}}$

So, $h = -\frac{1}{2}$

Then $\sigma_d = \frac{1}{|h|}\sigma_n = 2 \times 3.5 = 7$

74. C

Sol. We have for n = 1, $S_1 = T_1$

$\Rightarrow T_1 = (1 + 2T_1)(1 - T_1)$

$\Rightarrow T_1 = \frac{1}{\sqrt{2}}$

$\Rightarrow 2T_1^2 = 1$

75. B

Sol. $\log_2\left(1 + \sqrt{6x - x^2 - 8}\right) \geq 0$

$\Rightarrow 1 + \sqrt{6x - x^2 - 8} \geq 1 \Rightarrow 6x - x^2 - 8 \geq 0$

$\Rightarrow x^2 - 6x + 8 \leq 0$

$\Rightarrow (x - 2)(x - 4) \leq 0$

$\Rightarrow 2 \leq x \leq 4.$

Now $f'(x) = x^2 + 2x + 2 > 0 \ \forall \ x \in R$

$\Rightarrow$ f(x) is strictly increasing in [2, 4]

$f(x) = \frac{x^3}{3} + x^2 + 2x$

$\alpha = f(2) = \frac{8}{3} + 4 + 4 = \frac{32}{3}$

$\beta = f(4) = \frac{64}{3} + 16 + 8 = \frac{136}{3}$

$|\alpha - \beta|max = \frac{104}{3}.$

76. A

Sol. $E_1 \Rightarrow$ Person is accident person

$E_2 \Rightarrow$ Person is not accident person

A = person will have an accident with in a year

$P(A) = P(E_1)\ P(A/E_1) + P(E_2)P(A/E_2) = 0.3 \times 0.6 + 0.7 \times 0.8 = 0.74$

77. C

Sol. Let $\frac{1}{H_{i+1}} - \frac{1}{H_i} = k$

$$\sum_{i=1}^{2n}(-1)^i\left(\frac{H_i+H_{i+1}}{H_i-H_{i+1}}\right)=\sum_{i=1}^{2n}\frac{(-1)^j}{k}\left(\frac{1}{H_{i+1}}+\frac{1}{H_i}\right)=2n$$

78. D

Sol. $-1\le x^2-10x+26\le 1 \Rightarrow x = 5$ only and at $x = 5$

$\frac{\pi}{2}=0$ which not possible. Hence $a \in \phi$.

79. A

Sol. $z=\begin{vmatrix}\omega & \omega^2 & \omega\\ 2i & -3i & 7i\\ \omega^2 & \omega & \omega^2\end{vmatrix}$ $R_1 \to (R_1 + R_3) = \begin{vmatrix}-1 & -1 & -1\\ 2i & -3i & 7i\\ \omega^2 & \omega & \omega^2\end{vmatrix}$

$= -1(-3\omega^2 i-7\omega i)+(2\omega^2 i-7\omega^2 i) \Rightarrow -1(2\omega i+3\omega^2 i)=5i(\omega-\omega^2)=-5\sqrt{2}$

80. D

Sol. $f'(x) = 6x^2 + 2\alpha x + \beta$

$3\cdot 3^t=-\frac{2\alpha}{6}=-\frac{\alpha}{3}$

$3^{(t+1)}=-\frac{\alpha}{3}$

$f''(x)=12x+2\alpha=0$ at $x=\frac{5}{2}$ $\Rightarrow -\alpha=\frac{12\cdot(5/2)}{2}=15$

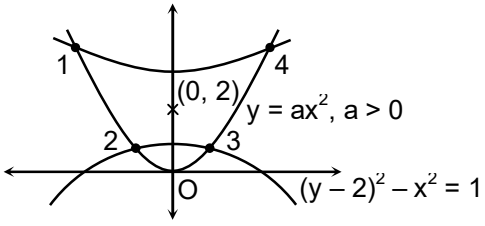
$\Rightarrow 3^{(t+1)}=5 \Rightarrow (t+1)=\log_3 5$ $\Rightarrow t=\log_3\left(\frac{5}{3}\right)$

SECTION – B

81. 00004.00

Sol. $(y-2)^2-x^2-1$

$y = ax^2$



82. 00003.00

Sol. Clearly $\sin A + 2\sin 2B + 3\sin 3C = 4$

$\Rightarrow \sin^2 A+\sin^2 2B+\sin^2 3C\ge\frac{4^2}{14}$

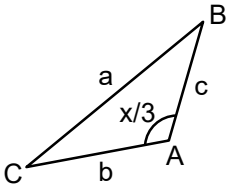
83. 00013.00

Sol. $a + b = 20$, $a + c = 21$

$\cos A=\left(\frac{b^2+c^2-a^2}{2bc}\right)$

$\Rightarrow 2a^2-123a+1261=0$

$\Rightarrow a = 13$



84. 00008.00

Sol. Let X = x – 3, Y = y – 1

$\Rightarrow |X + Y| = 5, |X| + |Y| = 5$

$\Rightarrow |X + Y| = |X| + |Y| \Rightarrow X.Y \geq 0$

Hence number of points are 8

85. 00002.50

Sol. D.R's of normal to plane x + y + z – 1 = 0 and x + ky + 3z – 1 = 0 is (1, 1, 1) and (1, k, 3) respectively

$\Rightarrow$ D.R. of normal to a plane perpendicular to given planes

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 1 & k & 3 \end{vmatrix} = \hat{i}(3-k) - \hat{j}(2) + \hat{k}(k-1)$$

$$\Rightarrow \frac{1+2\lambda}{3-k} = \frac{2+\lambda}{-2} = \frac{1+3\lambda}{k-1} \Rightarrow -2 - 4\lambda = 6 + 3\lambda - 2k - \lambda k \text{ and } -2 - 6\lambda = -2 - \lambda + 2k + \lambda k$$

$-4 - 10\lambda = 4 + 2\lambda \Rightarrow 12\lambda = -8 \Rightarrow \lambda = -\frac{2}{3}$

$$\Rightarrow -2 + \frac{8}{3} = 6 - 2 - 2k + 2 - k = 3 \Rightarrow \frac{2}{3} - 4 = -\frac{4}{3}k \Rightarrow -\frac{10}{3} = -\frac{4}{3}k \Rightarrow k = \frac{5}{2} = 2.5$$

86. 00054.00

Sol. Here $x_1x_2x_3 = 2^2 \times 3 \times 5$

Let number of two's given to each of x_1 , x_2 , x_3 be a, b, c .

Then a + b + c = 2, a, b, c ≥ 0

The number of integral solutions of this equations is equal to coefficient of x^2 in $(1 - x)^{-3}$ i.e. ${}^4C_2$

i.e. the available 2 two's can be distributed among x_1, x_2 and x_3 in ${}^4C_2$= 6 ways.

Similarly, the available 1 three can be distributed among x_1, x_2, x_3 in ${}^3C_2$ = 3 ways. (= coefficient of x in $(1 - x)^{-3}$)

$\therefore$ Total number of ways = ${}^4C_2 \times {}^3C_2 \times {}^3C_2 = 6 \times 3 \times 3 = 54$ ways.

87. 00000.50

Sol. Given R cos A = 2R cos B cos C = R [cos (B + C) + cos (B – C)]

$\Rightarrow$ 2 cos A = cos (B – C)

88. 00024.00

Sol. F(x) = f(x).g(x).h(x)

$\Rightarrow F'(x) = f'(x).g(x).h(x) + f(x).g'(x).h(x) + f(x).g(x).h'(x)$

$\Rightarrow F'(x_1) = f'(x_1).g(x_1).h(x_1) + f(x_1).g'(x_1).h(x_1) + f(x_1).g(x_1).h'(x_1)$

$\Rightarrow 21F(x_1) = 4\, f(x_1).g(x_1).h(x_1) + (-7)\, f(x_1).g(x_1).h(x_1) + k\, f(x_1).g(x_1).h(x_1)$

$\Rightarrow 21\, F(x_1) = (4 - 7 + k)\, F(x_1)$

$\Rightarrow k = 24.$

89. 00003.50

Sol. Let eccentricity of ellipse and hyperbola be e_1 and e_2 respectively, then

$b^2 = 16\left(1 - e_1^2\right)$ and $81 = 144\left(e_2^2 - 1\right)$

$$\Rightarrow e_1^2 = 1 - \frac{b^2}{16} \text{ and } e_2^2 = \frac{81}{144} + 1 = \frac{225}{144} \Rightarrow e_2 = \frac{15}{12}$$

Now given $4e_1 = \frac{12}{5}e_2 \Rightarrow 4\sqrt{1-\frac{b^2}{16}} = \frac{12}{5}\times\frac{15}{12} = 3$

$1-\frac{b^2}{16} = \frac{9}{16} \Rightarrow b^2 = \frac{7}{16}\times 16 \Rightarrow b^2 = 7$.

Latus rectum of ellipse = $\frac{2b^2}{a} = \frac{2\times 7}{4} = \frac{7}{2}$.

90. 00000.20

Sol. $\cot^2 36^\circ \cot^2 72^\circ = (\cot^2 36^\circ \cot^2 72^\circ - 1) + 1$

$$= \left(\frac{\cos^2 36^0 \cos^2 72^0 - \sin^2 36^0 \sin^2 72^0}{\sin^2 36^0 \sin^2 72^0}\right) + 1$$

$$= \left(\frac{\left(1+\cos 144^0\right)\left(1+\cos 72^0\right) - \left(1-\cos 144^0\right)\left(1-\cos 72^0\right)}{4\sin^2 36^0 \sin^2 72^0}\right) + 1 = \frac{\cos 144^0 + \cos 72^0}{2\sin^2 36^0 \sin^2 72^0} + 1$$

$$= \frac{\cos 108^0 \cos 36^0}{\sin^2 36^0 \sin^2 72^0} + 1 = \frac{-\cos 72^0 \cos 36^0}{\sin^2 36^0 \sin^2 72^0} + 1 = 1 - \frac{\left(\cot 36^0 \cot 72^0\right)^2}{\cos 36^0 \cos 72^0}$$

$$= 1 - \frac{2\sin 36^0 \left(\cot 36^0 \cot 72^0\right)^2}{2\sin 36^0 \cos 36^0 \cos 72^0} = 1 - \frac{4\sin 36^0 \left(\cot 36^0 \cot 72^0\right)^2}{\sin 144^0}$$

$\Rightarrow \cot^2 36^\circ \cot^2 72^\circ = 1 - 4\cot^2 36^\circ \cot^2 72^\circ$

$\therefore 5\cot^2 36^\circ \cot^2 72^\circ = 1$

$\Rightarrow \cot^2 36^\circ \cot^2 72^\circ = \frac{1}{5}$.